

Lecture 27: Vector fields in 3D: surface integrals & flux

Vector fields in space: At every point, $\vec{F} = \langle P, Q, R \rangle$
functions of (x, y, z)

Examples: * force fields (e.g.: gravitational attraction of a solid mass at the $(0,0,0)$ on a mass m at $(x, y, z) = \vec{F}$ directed towards origin)
 magnitude $\sim \frac{C}{r^2}$

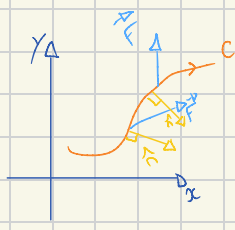
$$\vec{F} = -C \frac{\langle x, y, z \rangle}{r^3}$$

* electric fields, magnetic fields,

* velocity fields $\vec{v}$

* gradient fields $u = u(x, y, z)$
 $\nabla = \langle u_x, u_y, u_z \rangle$

Flux: in 2D
 $\int_C \vec{F} \cdot \hat{n} ds$



In 3D, flux is through a surface; surface integral

$\vec{F}$ vector field, S surface in space

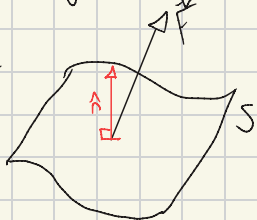
$\hat{n}$ = unit normal vector to

2 choices for $\hat{n}$! Choose a side of ("orientation") the surface

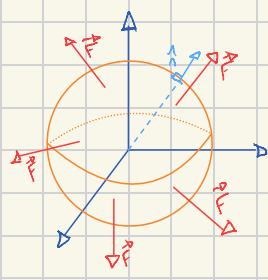
$$\text{Flux} = \iint_S \vec{F} \cdot \hat{n} dS$$

surface area element

Notation: $d\vec{S} = \hat{n} dS$



Examples: 1) Flux of $\vec{F} = \langle x, y, z \rangle$ through sphere of radius a at origin



$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n} dS$$

$$\hat{n} = \frac{1}{a} \langle x, y, z \rangle$$

(unit since $\sqrt{x^2 + y^2 + z^2} = a$)

$$\vec{F} \cdot \hat{n} = \frac{\vec{F} \cdot \vec{F}}{a} \text{ since } \vec{F} \parallel \hat{n}, \text{ so } \vec{F} \cdot \hat{n} = a$$

$$\iint_S \vec{F} \cdot \hat{n} dS = \iint_S a dS = a \left[\iint_S dS \right] = a \cdot \text{area}(S) = 4\pi a^2$$

2) Same sphere, $\vec{H} = z\hat{k}$

$$\text{We still have } \hat{n} = \frac{\langle x, y, z \rangle}{a} \Rightarrow \vec{H} \cdot \hat{n} = \langle 0, 0, z \rangle \cdot \frac{\langle x, y, z \rangle}{a} \Rightarrow \vec{H} \cdot \hat{n} = \frac{z^2}{a}$$

$$\iint_S \vec{H} \cdot \hat{n} dS = \iint_S \frac{z^2}{a} dS \quad dS = ?? d\phi d\theta$$

$$\Delta S \approx (a \sin \phi \Delta \theta) \cdot (a \Delta \phi)$$

$$dS = a^2 \sin \phi d\phi d\theta$$

$$z = a \cos \phi$$

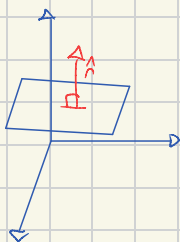
$$\iint_S \frac{z^2}{a} dS = \int_0^{2\pi} \int_0^\pi \frac{a^2 \cos^2 \phi}{a} \cdot a^2 \sin \phi d\phi d\theta$$

$$= 2\pi a^3 \cdot \left[-\frac{1}{3} \cos^3 \phi \right]_0^\pi = \frac{4}{3} \pi a^3$$

Conclusion for 2 examples: use geometry or set up $\iint_S \vec{F} \cdot \hat{n} dS$

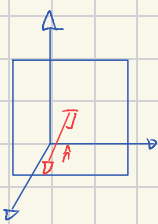
1) S = Horizontal plane $z = a$

$$\hat{n} = \pm \hat{k}, dS = dx dy$$



Vertical plane // yz-plane $x = a$

$$\hat{n} = \pm \hat{i}, dS = dy dz$$



1) sphere of radius a centered at origin

$$\hat{n} = \pm \frac{\langle x, y, z \rangle}{a} \quad dS = a^2 \sin \phi d\phi d\theta$$

2) cylinder of radius a centered around z -axis

$$\hat{n} = \pm \frac{\langle x, y, 0 \rangle}{a} \quad dS = a dz d\theta$$

3) Graph $z = f(x, y)$

$$\hat{n} dS = \pm \underbrace{\langle -f_x, -f_y, 1 \rangle}_{\text{not } \hat{n}} \underbrace{dx dy}_{\text{not } dS}$$

To setup bounds on $\iint \dots dx dy$ look at shadow of S in xy -plane.

Geometric Interpretation:

If $\vec{F}$ is a velocity field, Flux = amount of matter that passes through S per unit time.